



Indian Association for the Cultivation of Science
(Deemed to be University under *de novo* Category)
Master's/Integrated Master's-PhD Program/Integrated
Bachelor's-Master's
Program/PhD Course
End-Semester Examination-Autumn 2025

Subject: SEC
Full Marks: 50

Subject Code(s): 2101
Time Allotted: 3 hrs

Answer any five of the following questions each of 10 marks
Symbols have their usual meaning

1. (a) Distinguish between the following cases with real-life examples:
(i) primary and secondary data,
(ii) attribute and variable,
(iii) nominal and ordinal data,
(iv) discrete and continuous variables.
(b) When a pie chart is used?
(c) Describe a histogram and comment on its use to approximate the underlying distribution curve.
2. (a) Define the Karl Pearson's coefficient of correlation.
(b) Visually represent the cases of positive, negative and no correlation.
(c) Show that zero correlation between two variables does not guarantee independence.
(d) Prove $-1 \leq r_{xy} \leq 1$. Interpret the terminal cases.
(e) Describe the coefficient of determination.
3. (a) Define the two regression lines in a bivariate set-up.
(b) Show that the lines intersect at the respective means of the two variables under study.
(c) Discuss the scenario when the two lines coincide.
(d) If three uncorrelated variables x_1, x_2, x_3 have the same standard deviation, find $r_{(x_1+x_2)(x_2+x_3)}$.

Please Turn Over

- (c) If two variables x and y are linearly related as $y = a + bx$, where a and b are constants, show that $r_{xy} = 1$ or -1 as b is positive or negative.
4. (a) Derive the mode of a binomial distribution with parameters (n, p) .
 (b) Show that Poisson distribution can be derived as a limiting form of a binomial distribution.
 (c) Find the points of inflection for the normal curve with mean μ and standard deviation σ .
5. (a) Show that the population mean and variance are estimated without bias by their respective sample versions.
 (b) Define the following terms each with an example of hypothesis testing problem:
 (i) null hypothesis,
 (ii) alternative hypothesis,
 (iii) type I error,
 (iv) type II error,
 (v) p -value.
 (c) Define the term 'Pearsonian χ^2 ' in the context of testing of hypothesis.
6. (a) We have the following values for 25 pairs of observations on two variables x and y :
 $n = 25, \sum x = 125, \sum x^2 = 650, \sum y = 100, \sum y^2 = 460, \sum xy = 508$.
 (i) Compute the value of the correlation coefficient between x and y .
 (ii) If two pairs of values (8,12) and (6,8) were recorded mistakenly as (6,14) and (8,6) respectively, find the correct value of the correlation coefficient.
 (b) Given two regression equations as $x + 2y - 5 = 0$ and $2x + 3y - 8 = 0$
 (i) find the regression line of x on y .
 (ii) Find the means of x and y from the regression equations.
 (iii) Find the variance of y for given the variance of x as 12.
 (c) Briefly outline the student's t -test.